

Foundational Research & Advances in Quantum Matrix Inversion & Conditioning

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1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for Quantum Matrix Inversion & Conditioning in modern quantum information architectures.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"If a matrix is wobbly and distorted, inverting it amplifies tiny noise into giant errors. Conditioning smooths out the matrix so the quantum solver runs smoothly."

3. Mathematical Formulation & Unitary Rule

$$\kappa(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}, \quad P_{\text{success}} \propto \frac{1}{\kappa(A)^2}$$

The condition number determines both post-selection success and required precision in quantum linear solvers.

5. Executable IBM Qiskit (Python) Code

```
import numpy as np

A = np.array([[4.0, 1.0], [1.0, 3.0]])
kappa = np.linalg.cond(A)
print(f"Condition number kappa: {kappa:.2f} (well-conditioned for HHL)")
```

6. Computational Complexity & Speedup

Classical Time:	Quantum Time:	Speedup Class:
$O(N^{2.37})$ via Coppersmith-Winograd	$O(p \log N \log(1/\epsilon))$ (post-selection)	Exponential hardware

7. Key Applications & Future Research Directions

- Big-data covariance matrix inversion
- Optimal control in robotics and avionics
- Accelerating training of Gaussian Process regression models

Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant LinearAlgebra pipelines.